

Atomic AI - System Architecture

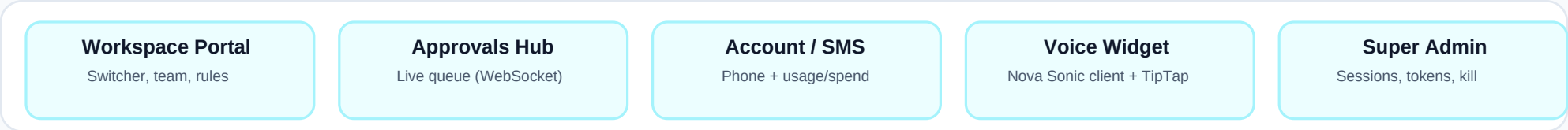
Multi-tenant AI automation platform | FastAPI (Python 3.14) + Next.js 16 | Strands + Amazon Bedrock | Docker Compose

CLIENTS



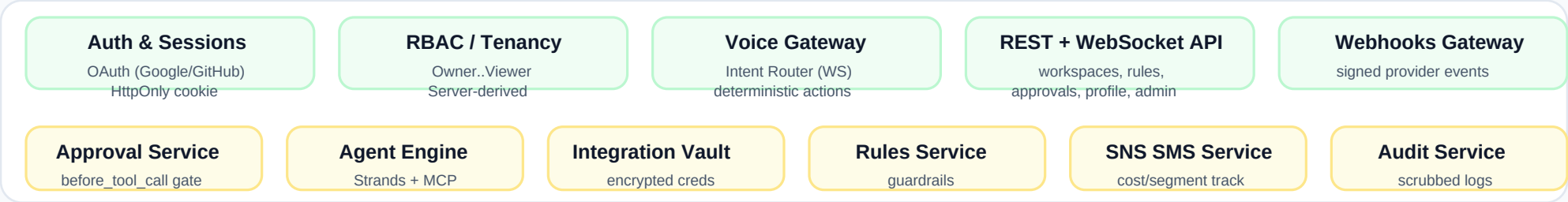
FRONTEND - Next.js 16 (App Router, TypeScript, Tailwind)

HTTPS / WSS



BACKEND - FastAPI (Python 3.14) / Uvicorn

authenticated API + WebSocket



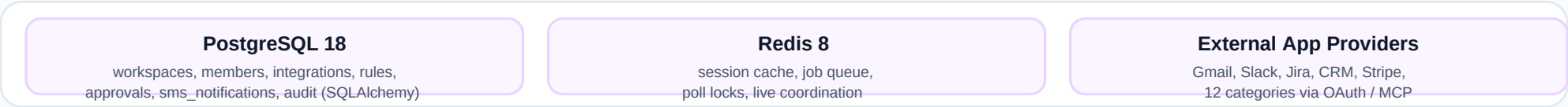
AGENT RUNTIME & AWS (shared credential chain + region)

run / notify



DATA & STATE

read / write



Human-in-the-loop: a high-impact tool call is paused by the Approval Service, posted to the Approvals Hub (live), and only executes after an Owner/Admin approves; every action is audited.